SUTRA — High-Level Design (Technical Proposal)

Statewide Unified Tracking, Registry & Analytics Gujarat Police CCTV Integration Hackathon 2026 · Category 1 (Individual) · Lavesh Paryani Working platform: https://github.com/laveshparyani/SUTRA (all open-source)

---

1. Solution model and justification

Hybrid: Model 1 (Registry & GIS Foundation) + Model 3 (Federation Middleware), with metadata-first analytics in the spirit of Model 2 and selective Model 4 capabilities (central analytics, tiered evidence storage) where they earn their cost.

Why this hybrid:

- **Model 1 is mandatory** and is genuinely load-bearing here: every downstream function (scheduling, alerts, route reconstruction, gap analysis) keys off registry metadata.
- **Model 3 over Model 2** because 26 departments will never converge on one VMS: an adapter/connector layer that federates *sources* (RTSP, ONVIF, HTTP/HLS, vendor SDKs, files) and *databases* (VAHAN, eGujCop…) behind stable interfaces is the only architecture that survives vendor churn. Departments keep their infrastructure and control.
- **Not Model 4** as the primary posture: centralising 80,000 video streams is a bandwidth and cost trap (see section 8). SUTRA centralises *metadata and evidence*, not video. Recording stays at the department/edge tier; the centre stores detections, alerts, thumbnails and audit — kilobytes per camera-minute instead of megabits per second.

2. Architecture

```
+---------------------------------------------------------------------+
|                     SUTRA COMMAND  (React + Leaflet)                |
|  overview map · video wall · vehicle trace · alerts · registry ·    |
|  coverage/gap analysis · watchlist · audit                          |
+--------------^-----------------------------^-----------------------+
               | REST (JWT, RBAC)            | WebSocket (alerts)
+--------------+-----------+  +--------------+-----------------------+
|      SUTRA ATLAS         |  |            SUTRA WATCH               |
| registry · GIS · health  |  | watchlist · fuzzy matcher · alerts   |
| gap/ageing analysis ·    |  | gov-DB connectors (VAHAN/SARTHI/     |
| bulk/API onboarding ·    |  | eGujCop/AFIS — adapter interface)    |
| CSV export · audit trail |  +--------------^-----------------------+
+--------------^-----------+                 | detections
               | metadata    +---------------+-----------------------+
               |             |            SUTRA INSIGHT              |
               |             | ANPR: plate det (YOLOv9-t ONNX) ->    |
               |             | OCR (CCT ONNX) -> temporal voting ->  |
               |             | Indian-plate normalisation/state-code |
               |             | repair · scene analytics (YOLOX-nano) |
               |             | · route reconstruction                |
               |             +---------------^-----------------------+
               |                             | sampled frames
+--------------+-----------------------------+-----------------------+
|                        SUTRA BRIDGE                                 |
| source adapters: http-progressive · RTSP(TCP) · file · HLS/ONVIF    |
| adaptive ingest scheduler (concurrency budget, dwell rotation,      |
| pinning, alert-boost) · health probes · MJPEG relay (shared)        |
+--------------^-----------------------------------------------------+
               |
   26 departments' cameras/VMS/NVRs · private/society feeds (RTSP)
```

Verified in the working platform (not proposed — running): every box above exists and was demonstrated on the hackathon portal's live feeds and local footage. Detection at 30–75 ms/frame and scene analytics at 43 ms/frame on a single CPU (no GPU present in the build machine).

3. Heterogeneous integration approach

- **Source adapters.** One ingest worker abstraction over FFmpeg (`http-progressive`, `rtsp` — forced TCP transport after field-testing showed RTP-over-UDP silently dropped by firewalls — `file`, HLS-ready). Container and codec heterogeneity (mp4/mkv/avi, H.264/HEVC) is handled by the decode layer; workers discard mid-GOP joins (HEVC corruption), auto-reconnect with backoff, and report per-camera health honestly (`connecting` is not `ok`; `last_frame_at` advances only on decoded frames).
- **VMS federation.** Vendor VMS/NVR systems integrate through the same adapter contract (discover / probe / open_stream / health); ONVIF Profile S/T and vendor SDK adapters are additive modules, no core change. The hackathon dataset itself carries Genetec VMS provenance — consumed without any Genetec software.
- **Onboarding paths (Model 1):** portal API auto-discovery, single-camera API, bulk CSV (tested with private/society cameras), manual UI. All audited.

4. Geographic dispersion and the ingest scheduler

Statewide links are constrained and shared. SUTRA treats stream concurrency as a budgeted resource:

- **Adaptive ingest scheduler:** live connections are capped by a per-node budget; cameras beyond it rotate through slots (dwell timer, least-recently-served fairness, staggered connects). Operators **pin** cameras; a watchlist hit **boosts** the alert camera *and its nearest neighbours* to resident slots — coverage tightens around a sighting instead of rotating away from it.
- Field-measured on the hackathon portal (which sustains ~8–10 concurrent streams): budget respected, rotation verified, boost verified end-to-end.
- Viewing is relayed (one ingest connection serves all viewers via MJPEG/WebRTC relay), never fanned out to sources.

5. AI-powered video analytics

Capability	Implementation (all open-source, CPU-proven)
Plate detection	YOLOv9-tiny 640 (ONNX), 8 MB
Plate OCR	CCT-S v2 (ONNX), 5 MB, per-character probabilities
Accuracy layer	**Temporal voting** (per-vehicle track, char-level probability vote) + **Indian-plate normalisation** (structure-aware confusion repair, state-code validation against the official code list) + fuzzy watchlist matching (confusion-fold + edit distance)
Scene analytics	YOLOX-nano (ONNX): person/vehicle counts per camera (crowd and intrusion rules build on this)
Route reconstruction	Detections grouped into sightings per camera, time-ordered, rendered as GIS polyline + timeline with evidence
FRS (roadmap)	Same sidecar pattern: face detection (SCRFD) + embedding (ArcFace ONNX) matched against eGujCop/AFIS-fed galleries; flagged-candidate → human-confirmation workflow, never automatic identification

Honest performance characterisation from real footage: close-range cameras (toll gates, showroom approaches) read at 0.81–0.98 confidence; wide-angle intersection PTZs yield partial reads (50–90 px plates) that the voting and fuzzy layers absorb. Two-line commercial plates produce OCR variants — the watchlist supports investigator-registered variants under one FIR.

6. Watchlist correlation and alerting

- Representative watchlist DB (stolen / blacklisted / suspect / wanted) with FIR references and priorities; production sources map 1:1 from eGujCop/CCTNS.
- Every finalised read is matched (exact → entry's priority severity; fuzzy "probable" → one severity lower). Alert → DB row + **WebSocket push** to all operators (toast + alert centre), annotated evidence frame, VAHAN-connector vehicle details, acknowledge workflow, 15-minute per-plate/camera cooldown, full audit.
- An alert also triggers the scheduler boost (section 4) — the system reacts operationally, not just visually.

7. Cybersecurity, privacy, auditability

Implemented and covered by an automated security test suite (19 tests: auth, RBAC, media protection, traversal, input validation):

- JWT auth (PBKDF2, 200k iterations); per-install auto-generated signing secret (never a committed default); login **rate limiting** (sliding window per client, audited failures); RBAC: admin / department operator (department-scoped data visibility, enforced server-side) / viewer.

- **All media protected**: camera snapshots, MJPEG streams, evidence images and the alert WebSocket authenticate via an HttpOnly, SameSite cookie (invisible to page scripts — XSS cannot exfiltrate it); evidence serving is confined to the data directory with URL-encoding-safe path resolution and an image-type allowlist.
- **Input validation**: camera source URLs restricted to camera protocols; file sources confined to the media directory (no SSRF into internal services, no arbitrary server-file reads); all user-visible strings HTML-escaped (map popups included); ORM-parameterised queries throughout.
- Security headers (nosniff, frame-deny, no-referrer); CORS restricted to configured origins; audit trail of logins (including failures), onboarding, exports, watchlist changes, acknowledgements — surfaced in the UI.
- Privacy by design: no continuous central video recording; watchlist-match-only alerting; evidence retention policy; owner names masked in connector responses.
- Production additions (documented): TLS termination, mTLS on federation links, secrets vault, network segmentation (ingest DMZ / analytics / data planes), SIEM export, hash-chained audit log.

8. Scalability to ~80,000 cameras

Design principle: video stays at the edge; metadata flows up.

- **Edge tier (district/zone, ~150 nodes):** ingest + ANPR on commodity servers. Measured: 30–75 ms/frame/core CPU-only ⇒ a 32-core node handles **300–500 cameras at 1 fps sampling**; one mid-range GPU (T4-class) raises that to ~1,500–2,500. 80k cameras ⇒ ~160–260 CPU nodes *or* ~40–60 GPU nodes.
- **Uplink per camera ≈ 2–6 KB/s** (detections + thumbnails) versus 2–4 Mb/s for video — a **~500× bandwidth reduction**; a district of 500 cameras federates over ~25 Mb/s. Full-video recall stays departmental (existing NVR retention: 7–15+ days as today).
- **Regional tier (6–8):** Kafka event bus, PostgreSQL + PostGIS (partitioned / Timescale), S3-compatible object store for evidence (hot 30 d / warm 180 d / cold archive per retention policy).
- **Central tier:** federation control plane, registry, search, command centre, gov-DB connectors. Kubernetes; horizontal scale-out; HA per tier; DR by cross-region replication of metadata + evidence (video DR remains departmental).
- **Load and health:** the prototype's scheduler/health model is the same control loop that governs edge nodes at scale; Prometheus/Grafana monitoring, structured logging, per-camera health as first-class data.
- **Indicative cost (edge CPU option):** ~₹2.4–3.6 L/node × 160–260 nodes ≈ **₹40–75 Cr capex** + network; GPU option ≈ ₹55–90 Cr with 4× analytics headroom — an order of magnitude under centralised-video designs, because the state pays for inference, not for hauling video.

9. Department prerequisites (integration feasibility)

Per department: camera/NVR inventory with RTSP/ONVIF endpoints or VMS API + credentials; network reachability to the district edge node (or a 4G/leased uplink); storage/retention declaration for the registry; a nodal officer for credential rotation. For databases: NIC/SCRB API access to VAHAN / SARTHI / eGujCop with IP allow-listing (connector contract in docs/CONNECTORS.md).

10. Roadmap

- **Phase 1 (now):** this platform.
- **Phase 2:** FRS + AFIS confirmation loop, ONVIF auto-discovery, Kafka bus, PostGIS migration.
- **Phase 3:** statewide rollout per section 8 tiers, eGujCop-fed watchlists, public/private feed onboarding portal.